Yinhan Lu

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EDUCATION

McGill University, Montreal, Quebec — **Aug 2023 - Apr 2026 (Expected)**

- Bachelor of Science, Honour Computer Science
- GPA: 3.93/4.00. Computer Science Courses: 4.00/4.00
- Dean's Honour List (2024-2025)
- Key Courses: Machine Learning, Reinforcement Learning, Natural Language Processing, Brain-Inspired AI, Data Structures & Algorithms, Operating Systems, Computer Systems, Linear Algebra, Probability, Statistics

EXPERIENCE

Student Researcher, McGill University & Mila - Quebec AI Institute — **Sept 2025 - Present**
Mentor: Eric Elmoznino — **Montreal, QC**

- Investigating novel sequence modeling architectures by implementing proposed models, training them on NLP benchmark datasets, and conducting systematic empirical evaluation to characterize performance trade-offs and identify conditions under which architectural innovations improve upon existing baselines.

Research Intern, McGill University & Mila - Quebec AI Institute — **Aug 2025 - Present**
Supervisor: David Adelani — **Montreal, QC**

- Developing computational methods for digitizing and structuring low-resource African language dictionaries and grammar books, establishing machine-readable lexical resources to enable NLP research in extremely low-resource language settings.
- Investigating approaches to incorporate structured lexical knowledge into machine translation systems for languages with minimal parallel corpora, examining how dictionary-based resources can improve model performance when traditional data-driven methods are insufficient.

AI Researcher, McGill AI Society Research Incubator — **Montreal, QC**
March 2025 - July 2025

- Explored bias detection methods in parliamentary discourse by training RoBERTa models from scratch on Inuit Parliament and Canadian Parliament datasets, implementing custom tokenizers and analyzing embedding space geometry to identify potential linguistic bias patterns.
- Investigated embedding-based approaches to bias detection as alternative to fine-tuning methods, examining how representation learning on demographically distinct corpora can reveal systematic differences in language modeling.

Research Intern, Université de Montréal & Mila - Quebec AI Institute — **May 2024 - Present**
Supervisor: Bang Liu, Mentor: Haochen Shi — **Montreal, QC**

- Independently designed and implemented a modular LLM-based agent system for autonomous exploration in open-world environments, developing integrated architecture for adaptive decision-making in complex, open-ended tasks.
- Investigating methods to improve agent reasoning and planning capabilities in autonomous settings through self-supervised learning of internal representations, exploring how agents can optimize their decision-making processes without external supervision.
- Conducting literature review on skill learning in LLM-based agents and contributing to development of multi-agent benchmarks for evaluating collaborative decision-making capabilities.

Research Intern, National Taiwan University, HCI Lab — **Jun 2024 - Aug 2024**
Supervisor: Mike Chen, Mentor: Wei-Chung Su & Shun-Yu Wang — **Taipei, Taiwan**

- Contributed to RoomDreaming, an AI-assisted iterative design system investigating how multi-round refinement processes improve creative exploration and output quality compared to single-shot generation approaches in human-AI collaborative design.
- Participated in research design and execution of user studies including formative interviews (N=6) and summative evaluations (N=20), examining owner-designer communication patterns and evaluating iterative AI assistance against traditional design exploration methods.
- Conducted literature review on human-AI collaboration in creative domains and contributed to study design, interaction design, and system implementation. Research findings informed successful commercial deployment, demonstrating real-world applicability of iterative generation paradigms in design practice.

ADDITIONAL

- **Programming Languages:** Python, C, C++, Java, JavaScript, TypeScript, R, Bash, OCaml, MATLAB
- **ML/AI:** PyTorch, Transformers, Gym, scikit-learn, NumPy, Pandas, Matplotlib
- **Tools & Technologies:** Git, Docker, Jupyter, Hugging Face, Weights & Biases, React.js, Unix/Linux, LaTeX, Figma
- **Languages:** English (fluent), Mandarin (native)